

Energy-Conditioned 3-D Residual U-Nets with 2.5-D UNet++ Synthetic CT for Proton Dose Prediction from CT and MRI

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Abstract. We describe one method for the DoseRAD2026 proton-dose tasks on computed tomography (CT) and magnetic resonance imaging (MRI). Both tasks use the same 3.09-million-parameter, six-level residual 3-D U-Net on a beam’s-eye-view (BEV) grid of $288 \times 64 \times 64$ voxels at $2 \times 1 \times 1$ mm. Proton energy is encoded by feature-wise linear modulation of deep network stages. The final 80-epoch continuation used masked beam mean absolute error (MAE) plus integrated depth-dose (IDD) distance, AdamW at 10^{-4} , cosine annealing to 4×10^{-5} , BF16 mixed precision, an effective batch of four, and paired lateral flipping ($p = 0.25$). Phase epoch 79 minimized the six-patient cached-BEV validation objective at 0.010897 (MAE 0.006335; IDD 0.004561) and was retained as the single dose checkpoint. For MRI, one 26.08-million-parameter 2.5-D UNet++ with a ResNet-34 encoder first maps five adjacent MR slices to synthetic CT (sCT). It was fine-tuned using body-masked L1 plus 0.15 times a SAM-based perceptual loss, AdamW, and coherent spatial and MR-intensity augmentation. The best fold-1 checkpoint by validation body MAE was retained. At inference, complete slices are processed in FP16 batches of 16, converted to $[-1024, 3071]$ HU, and set to air wherever the original MR is exactly zero. Order-3 cubic B-splines map anatomy to BEV and dose back to the native grid. Predictions are non-negative and values at or below each beamlet’s provided minimum cut-off are zeroed; no upper clipping or ensemble is used. Native-space dose metrics must still be recomputed for the final checkpoint. Independent 12-patient sCT validation gave body MAE 51.995 ± 14.756 HU. A historical 176-beamlet HTTP profile of the same architecture required 7.912 s; sCT generation averaged 1.265 s.

Keywords: Proton dose calculation · residual U-Net · synthetic CT · magnetic resonance imaging · beam’s-eye view.

1 Introduction

Accurate dose calculation is central to radiotherapy planning, but reference Monte Carlo simulation remains computationally demanding. DoseRAD2026 provides paired CT and MRI from thoracic and abdominal cases together with

beam-level Monte Carlo dose maps for photon and proton irradiation, enabling controlled evaluation of learned dose engines [10]. The proton tasks are particularly sensitive to heterogeneous tissue composition because beam range depends on the material traversed before the Bragg peak. MRI adds a second difficulty: it provides valuable soft-tissue contrast but does not directly encode the electron-density information normally derived from CT.

We entered the proton-on-CT and proton-on-MRI task variants. Our design separates image-dependent geometry from beamlet-dependent inference. A compact energy-conditioned residual 3-D U-Net predicts each beamlet in a fixed BEV coordinate system. This reduces the spatial variability seen by the network and permits the same dose model to process all proton energies and both anatomical regions. MRI is handled by a preceding MR-to-sCT network rather than by retraining the dose model directly on MR. This modular choice reuses the same CT-calibrated dose mapping, while making the additional MR uncertainty and latency explicit.

Our contributions are a compact energy-conditioned BEV dose model shared by both tasks, a full-slice 2.5-D sCT route that avoids patch-stitching artifacts, and an inference implementation that reuses spline coefficients and projected anatomy across beamlets. We batch the 3-D network, perform cubic interpolation on the GPU, overlap output compression with GPU work, and generate each sCT only once. The final submission will use one dose and one sCT checkpoint; ensembles were rejected when image-quality gains did not improve dose. We report its training, internal validation, and runtime while distinguishing model-selection data from independent and diagnostic evaluations.

2 Methods

2.1 Data and validation splits

The DoseRAD2026 data comprise 75 public training patients and 40 withheld test patients, with 36 abdominal and 39 thoracic patients in the training cohort. Each training patient has paired, geometrically aligned CT and MR volumes. The proton subset contains 36 gantry angles, 15 rays per angle, and two sampled energies per ray, or 1,080 beamlets per patient and 81,000 beamlets in total. Native proton image and dose spacing is $1 \times 1 \times 3$ mm, and energies span 31.7290–200.7966 MeV [10]. We used no additional dose labels.

The final dose phase used 69 training patients and six model-selection patients (1ABB039, 1ABB042, 1ABB061, 1THB008, 1THB120, and 1THB221; 6,480 beamlets). Four cases held out during development (1ABB135, 1ABB149, 1THB037, and 1THB122) were added to final-phase training. Consequently, the earlier ten-patient native-space analysis is not a held-out evaluation of the final checkpoint.

The sCT fold used 63 training and 12 validation patients. Validation patients were selected as six representatives per anatomy by K-means followed by the nearest observed sample (medoid), using image dimensions, body/intensity

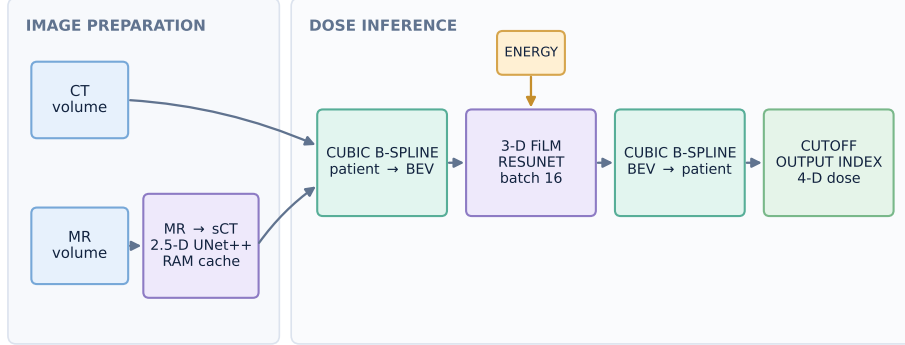


Fig. 1. Final CT and MRI inference pipelines. The two task variants share the same energy-conditioned dose model and patient–BEV transforms. The MRI branch adds one in-memory sCT generation per unique image; no intermediate image is written to disk.

statistics, and beam geometry. Fold-1 validation cases were disjoint from its 63 training cases. The initialization was the public ImpactSynth MR–CBCT CV-1 checkpoint, whose documented pretraining data include SynthRAD 2023 and SynthRAD 2025 Task 1 [1]; this is external pretraining and is reported separately from DoseRAD fine-tuning.

2.2 Beam-centered representation and dose network

For a ray from source $\mathbf{s}$ to target $\mathbf{t}$, we formed an orthonormal BEV frame from the propagation direction, a transverse lateral axis, and the patient superior axis. The anatomical image was transformed with an order-3 B-spline [7] to a grid of $288 \times 64 \times 64$ voxels (depth, lateral, superior) at $2 \times 1 \times 1$ mm. Its first depth plane lies 320 mm upstream from $\mathbf{t}$. Interpolated intensities were limited to the physical range $[-1024, \max(I)]$ HU, then normalized as

$$x = \frac{\text{clip}(I, -1024, 3000) + 1024}{4024}. \quad (1)$$

The array was quantized to float16 in the preprocessing cache and restored to float32 by the training loader. Energy was mapped linearly from $[31.7290, 200.7966]$ MeV to $[0, 1]$.

The dose network, ProtonFiLM-Lite, is a residual 3-D U-Net derived from the contracting/expanding design of U-Net [6]. Six feature levels with base width six yield channel widths 6, 12, 24, 48, 96, and 192. Each residual block contains two 3^3 convolutions, GroupNorm [9], SiLU, and an identity or 1^3 projected skip. The decoder uses trilinear upsampling, 1^3 channel reduction, concatenation with the encoder skip, and a residual block. Dropout3D of 0.05 is used at the bottleneck and deepest decoder block.

A shared two-layer energy multilayer perceptron produces an embedding. Small stage-specific heads apply residual feature-wise linear modulation [4], $f' =$

$f(1 + \gamma) + \beta$, at encoder levels 3–5, the bottleneck, and decoder levels 5–4. The FiLM heads were initialized to zero, hence to the identity. The final 1^3 head was initialized with standard deviation 10^{-3} and bias -8 ; Softplus guarantees a non-negative BEV output. Training included two auxiliary heads; after removing them, the deployed network has 3,094,511 trainable parameters.

2.3 MR-to-sCT model

The MRI branch uses a 2.5-D UNet++ [11] implemented with Segmentation Models PyTorch. Five adjacent axial MR slices form the input and the central CT slice is the target. The encoder is ResNet-34 [2]; the nested decoder and tanh head produce one normalized CT slice. The 26,084,881-parameter model was initialized from ImpactSynth CV-1.

For each patient, the minimum and 99.5th percentile were computed on the full MR volume, including background. Intensities were mapped to $[-1, 1]$, clipped, and stored as float16. Training crops were 320×320 , centered on the body bounding box for each nonempty slice. Two neighboring slices on either side provided context; indices were reflected at the first and last slice. Values outside the image were padded with -1 .

At inference, the fully convolutional network processes each complete native slice once. Height and width are padded once to a multiple of 32, and there is no spatial tiling, stride, overlap, Gaussian importance map, or stitching. Consequently, the Gaussian blending used in earlier experiments is not part of the final pipeline. The normalized volume is uploaded once, five-slice contexts are gathered on GPU, and slices are inferred in FP16 batches of 16. Predictions are mapped to HU, clipped to $[-1024, 3071]$, rounded to int16, and voxels at which the original MR equals exactly zero are set to -1024 HU. The MR and CT grids are already matched, so no sCT resampling is performed.

2.4 Training

Table 1 summarizes the recorded configurations. The dose target was multiplied globally by 10^5 before float16 caching; predictions were divided by the same constant at inference. Let g_b and p_b be the ground-truth and predicted dose for beamlet b , $D_b = \max g_b$, and $M_b = \{v : g_b(v) \geq 0.1D_b\}$. The masked beam loss was

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|M_b|D_b} \sum_{v \in M_b} |p_b(v) - g_b(v)|. \quad (2)$$

The IDD loss summed dose over both transverse BEV axes and computed the RMS difference of the depth curves after normalization by the maximum ground-truth IDD. The main loss was $\mathcal{L}_{\text{dose}} = \mathcal{L}_{\text{MAE}} + \mathcal{L}_{\text{IDD}}$. Auxiliary decoder heads used the same objective with weights 0.20 and 0.10 during training only; validation and inference used the main head.

The base dose model used AdamW [3] with initial learning rate 10^{-3} and weight decay 5×10^{-4} on convolutional and linear weights only. A cosine schedule

Table 1. Recorded configurations of the final dose phase and sCT model.

	Dose model	MR-to-sCT model
Architecture	Six-level FiLM residual 3-D U-Net, 3.09 M deployed parameters	2.5-D UNet++/ResNet-34, 26.08 M parameters
Training input	$288 \times 64 \times 64$, spacing $2 \times 1 \times 1$ mm	Five 320×320 slices
Split	Final phase: 69 train; 6 monitored validation	63 train; 12 representative validation
Loss	Masked beam MAE + IDD; auxiliary weights 0.20/0.10	Body L1 + 0.15 SAM perceptual
Optimizer	AdamW, LR 10^{-4} , WD 5×10^{-4} (weights only)	AdamW, LRs $10^{-5}/5 \times 10^{-5}/10^{-4}$, WD 10^{-3}
Schedule	80-epoch cosine to 4×10^{-5}	None; constant learning rates
Batch/precision	2, accumulation 2; BF16 AMP	32; FP16 AMP
Augmentation	Paired lateral flip, $p = 0.25$	Flip, joint affine, MR intensity/noise
Final checkpoint	Continuation epoch 79, minimum validation loss	Fold 1 epoch 6, minimum validation body MAE

targeted 2.5×10^{-5} after 400 epochs. The physical batch was two, with two-step gradient accumulation (effective batch four), BF16 automatic mixed precision, gradient-norm clipping at 1.0, and a deterministic cache without stochastic augmentation. The final 80-epoch continuation started from the preceding phase’s `last.pt`, retained AdamW moments, scaler, and RNG state, and used a fresh cosine schedule from 10^{-4} to 4×10^{-5} . Loss, batch, BF16, and clipping were unchanged; paired CT-dose lateral flipping used $p = 0.25$. Phase epoch 79 (stored as zero-based epoch 78) minimized the main-head validation loss and was saved as `best.pt`.

For sCT fine-tuning, the loss was

$$\mathcal{L}_{\text{sCT}} = \mathcal{L}_{1,\text{body}} + 0.15 \mathcal{L}_{\text{SAM}}, \quad (3)$$

where only L1 was masked and SAM-2.1-S-derived perceptual features [5] were compared over the complete patch. AdamW parameter groups used constant learning rates 10^{-5} , 5×10^{-5} , and 10^{-4} for encoder, decoder, and head, respectively, with weight decay 10^{-3} and no scheduler. Training used FP16 mixed precision, batch 32, 32,000 samples (1,000 iterations) per epoch, and up to 50 epochs without early stopping. Augmentation comprised horizontal flipping ($p = 0.5$); a joint affine transform ($p = 0.35$, $\pm 5^\circ$, ± 10 pixels); and MR-only intensity augmentation ($p = 0.5$) with gamma and gain in $[0.9, 1.1]$, bias in $[-0.08, 0.08]$, and Gaussian-noise standard deviation in $[0, 0.025]$. Vertical flips were disabled. The final fold-1 checkpoint was epoch 6, selected by the lowest mean per-patient validation body MAE.

2.5 Inference and output construction

The two networks were loaded once during container startup, moved to GPU, placed in evaluation mode, and warmed before the health endpoint became ready. Inference used PyTorch inference mode, channels-last layouts, and batch 16 for both dose and sCT. Dose used BF16 autocast when the GPU supported it and FP16 otherwise; sCT used FP16. CuPy executed the spline prefilter and affine transforms on GPU.

One spline prefilter was computed per anatomy. BEV anatomy was cached by exact ray source and target, so the two energies on a ray reused the same volume. For MRI, an LRU cache retained up to ten int16 sCTs in CPU RAM and reused an sCT across Invoke calls when the same MR image and geometry recurred. No temporary sCT files are created.

The predicted BEV dose was divided by 10^5 and inverse-resampled with an order-3 B-spline directly into a preallocated native-grid output buffer. Negative cubic ringing was removed. For each beamlet, the challenge-supplied `minimum_cutoff` field was applied inclusively, $p_b(v) = 0$ whenever $p_b(v) \leq c_b$. This cutoff is independent of the 10% ground-truth mask used by the evaluation metric. No upper dose clipping, smoothing, amplitude calibration, or connected-component filtering was used. Each frame was placed explicitly at its supplied output-file and frame indices; ten compressed 4-D MetaImage stacks were written, including placeholders for unused slots. Compression was performed concurrently with GPU computation.

3 Evaluation

Masked beam MAE in Eq. (2) matches the current Level-1 source. However, its IDD integrates along a fixed array axis (`beam_axis=0`), whereas the native-space report script calls an unavailable older direction-and-spacing IDD interface. Its raw CT/MR reports are also absent. The results below are therefore historical internal diagnostics pending recomputation with the retained challenge evaluator. Metrics give beamlets equal weight within each patient and patients equal weight overall.

The final checkpoint was selected by the minimum main-head composite loss on the six cached-BEV validation patients. This surrogate uses the same MAE and IDD definitions as training, but is not a native-space Level-1 evaluation. The historical native-space CT evaluation used all 1,080 beamlets from each of ten patients before four were moved into final training. The MRI diagnostic used seven evenly spaced angles for each of the six dose-validation patients, retaining all rays and energies (1,260 beamlets). These six also belonged to sCT training, so the MRI result is not an independent sCT evaluation.

For sCT, body MAE was calculated inside a CT-derived mask: $CT > -900$ HU, 3-D binary closing, largest connected component, and hole filling. The 12-patient fold-1 validation was evaluated on reconstructed full volumes. On the six dose patients we additionally computed PSNR with fixed 4,095-HU data

Table 2. Historical pre-continuation native-space dose results; IDD requires reconciliation with the current fixed-axis evaluator. Values are mean $\pm$ population SD over patient means; CI is the two-sided 95% Student- t interval.

Task	Metric	Mean $\pm$ SD	Median	95% CI	Patients/beamlets
CT	Masked beam MAE	.007958 $\pm$.000687	.007842	[.007440,.008476]	10/10,800
CT	IDD distance	.007785 $\pm$.001456	.007592	[.006687,.008882]	10/10,800
MR	Masked beam MAE	.011526 $\pm$.000931	.011903	[.010456,.012596]	6/1,260
MR	IDD distance	.012500 $\pm$.001413	.012505	[.010876,.014124]	6/1,260

range and slice-wise Gaussian SSIM [8] ($\sigma = 1.5$), averaged inside the CT body. We report mean, population standard deviation (matching the saved evaluator reports), median, and a two-sided 95% Student- t confidence interval over patient means where available. These intervals quantify sampling uncertainty of small internal cohorts, not external generalization.

To quantify the numerical cost of the coordinate transforms independently of the network, we also retained an identity round-trip diagnostic: native reference dose was projected to BEV, stored as float16, and reconstructed on its original patient grid before masked beam MAE was computed. The proton experiment used the final $288 \times 64 \times 64$, $2 \times 1 \times 1$ -mm grid and order-3 B-splines in both directions for all 10,800 beamlets of the historical ten-patient cohort. An earlier seven-control-point photon diagnostic used a $256 \times 200 \times 200$ isotropic 2-mm grid to compare cubic and linear interpolation. The photon experiment therefore evaluates interpolation order on a different grid; it is not an estimate of the final proton pipeline.

Runtime was reported through the actual HTTP Invoke path after model loading and warm-up. The final dose continuation ran on one NVIDIA RTX 6000 Ada GPU (48 GiB), with an Intel Xeon W5-3425 CPU and 502 GiB host RAM. The historical HTTP profiler artifact is absent, so its workstation meta-data cannot be reverified. The documented Docker runtime used PyTorch 2.9.1, CUDA 12.6, CuPy 13.6, and SimpleITK 2.5.4.

4 Results

4.1 Dose and sCT accuracy

The final continuation completed 80 epochs. Phase epoch 79 was selected as `best.pt`, with cached-BEV validation loss 0.0108966 (masked MAE 0.0063354; IDD 0.0045612). These values are model-selection surrogates, not native-space metrics. Table 2 therefore retains the epoch-372 native-space results only as a pre-continuation reference. They must not be interpreted as the final checkpoint’s accuracy, particularly because four of the ten CT cases subsequently entered final-phase training.

The historical sCT report for the independent 12-patient fold-1 validation set gave a full-volume body MAE of 51.995 ± 14.756 HU (median 48.914, 95% CI [42.203,61.788]); values ranged from 37.371 to 91.615 HU, with the largest error on thoracic patient 1THB031. On the six non-independent dose patients, the

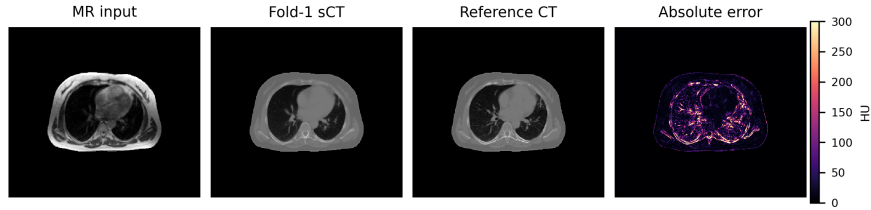


Fig. 2. Diagnostic example from thoracic patient 1THB008. MR display limits are its nonzero 1st–99th percentiles; CT and sCT use the same $[-1000, 1200]$ -HU window, and absolute error is clipped at 300 HU for display. This patient was in sCT fold-1 training and is shown qualitatively, not as an independent test case.

diagnostic body MAE, SSIM, and PSNR were 38.366 ± 3.831 HU, 0.869776 ± 0.016437 , and 34.396 ± 1.307 dB. A representative thoracic slice is shown in Fig. 2.

4.2 Patient–BEV round-trip diagnostic

The proton identity round-trip gave masked beam MAE 0.003264 ± 0.000200 (Table 3). This is not a strict lower bound on prediction error because a network may compensate for systematic effects. It jointly measures forward interpolation, float16 cache quantization, and inverse interpolation, whose contributions cannot be separated here. The ten-patient cohort predates the final continuation split; four of these patients subsequently entered training.

Table 3. Reference-dose round-trip masked beam MAE. Proton: mean $\pm$ SD of patient means; photon: median [minimum, maximum].

Data	Grid	Order	n	Masked beam MAE
Proton	$288 \times 64 \times 64$; $2 \times 1 \times 1$ mm	Cubic	10/10,800	$.003264 \pm .000200$
Photon	$256 \times 200 \times 200$; 2 mm	Cubic	7	$.006035$ [.003191,.011326]
Photon	$256 \times 200 \times 200$; 2 mm	Linear	7	$.017600$ [.012873,.034963]

On the photon controls, the median linear round-trip error was 2.92 times the cubic value. This diagnostic supports the retained order-3 interpolation, but the cross-grid comparison is not a paired proton ablation.

4.3 Single sCT model versus ensembles and postprocessing

In the pre-continuation diagnostic on six patients and 1,260 beamlets, three-fold averaging improved sCT MAE by 1.86% but worsened both dose metrics and approximately doubled sCT latency, so fold 1 alone was retained. Removing the exact-zero MR air mask or applying Gaussian smoothing worsened HU MAE; connected-component cleanup changed no voxels, while light dose smoothing had negligible and inconsistent effects. None of these postprocessings was retained.

4.4 Runtime

Table 4 reports a pre-continuation CT benchmark of the same architecture and batch size: one $590 \times 518 \times 146$ volume, 176 beamlets, and ten output stacks. Total Invoke time was 7.9118 s (45.0 ms per beamlet). BEV-to-patient interpolation remained the largest serial GPU/host stage. Compression CPU time totaled 7.1638 s but overlapped GPU work; writer waiting was 0.6605 s. Peak VRAM and process RSS were 6.461 and 2.663 GiB.

Across the six MRI volumes, loaded-and-warmed fold-1 sCT generation averaged 1.265 s (median 1.395, maximum 1.681) at batch 16. The model forward averaged 0.647 s; normalization and HU conversion averaged 0.344 and 0.165 s. The sCT cost is paid once per MR and amortized across its beamlets; a two-beamlet integration fixture was therefore not considered a representative full case.

Table 4. Historical local CT Invoke profile for 176 beamlets. Compression CPU time overlaps other stages and is shown separately from critical-path writer wait; the original profiler log must be restored for archival reproducibility.

Component	Time (s)
CT read	0.3133
Anatomy spline prefilter	0.2989
Anatomy-to-BEV projections	0.0916
Dose network (batch 16)	1.8819
BEV-to-patient interpolation	4.4145
Compression CPU work (overlapped)	7.1638
Writer wait on critical path	0.6605
Complete Invoke	7.9118

5 Discussion

The BEV representation lets a compact network learn range and lateral dose in a normalized geometry. FiLM supplies energy at deep levels while early image features remain shared across CT, MR, anatomies, and energies. Non-negativity and the supplied per-beamlet cutoff impose explicit output constraints without validation-fitted amplitude calibration.

The patient-BEV round-trip shows that interpolation is not lossless, while the photon diagnostic found substantially lower error with cubic than linear interpolation. This supports the retained order-3 transforms, without treating the round-trip value as a strict lower bound. sCT ensembling improved HU MAE but degraded dose and doubled latency, supporting the final single-model route.

All quantitative results are internal. Six CT patients were monitored during dose training, and all six MRI dose patients entered sCT fine-tuning, making the

end-to-end MR result optimistic. The independent sCT fold had higher MAE, including one 91.6-HU case. Moreover, public sCT pretraining used SynthRAD 2023/2025 data, while DoseRAD derives from SynthRAD2025.

The final dose checkpoint is continuation epoch 79, selected on six-patient cached-BEV loss. Its native-space results remain unavailable, so the dose table is explicitly a pre-continuation reference rather than final accuracy. Four CT cases in that table entered final training. A representative full MRI Invoke benchmark, raw reports, and beam/angle analysis are also unavailable.

Finally, the single-center cohort covers only thorax and abdomen with a standardized MR acquisition. Scanner shifts, metal artifacts, other anatomies, out-of-range energies, and clinical spot distributions remain untested. Hidden-cohort and independent MR dose evaluation are required before clinical use.

Author Contributions (CRediT)

Cédric Hémon: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing (original draft). **Valentin Boussot:** Methodology, Software, Writing (review & editing). **Jean-Claude Nunes:** Supervision, Writing (review & editing). **Caroline Lafond:** Supervision, Project administration, Writing (review & editing).

Disclosure of Interests. The authors have no competing interests to declare.

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